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## Day 2 Building Inns & Transfer Learning

### Learning objectives

- Booting ☒
- Full CNN Architecture ☒
- Data Augmentation ☒
- Transfer Learning ☒
- Mobile Net V2 ☒
- Freezing ☒
- Fine Tuning ☒

# 1. pooling

After conv2d filters are applied to the image, we have features maps but the problem that the feature maps can be very large

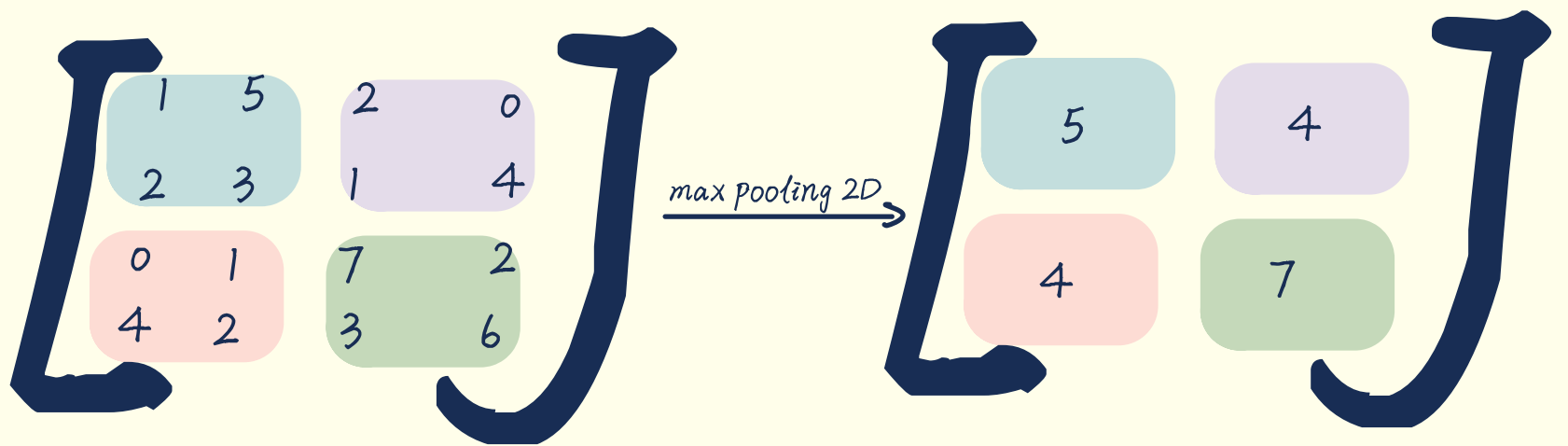
for Example

160 x 160 x 32

↳ So we use pooling to reduce the size

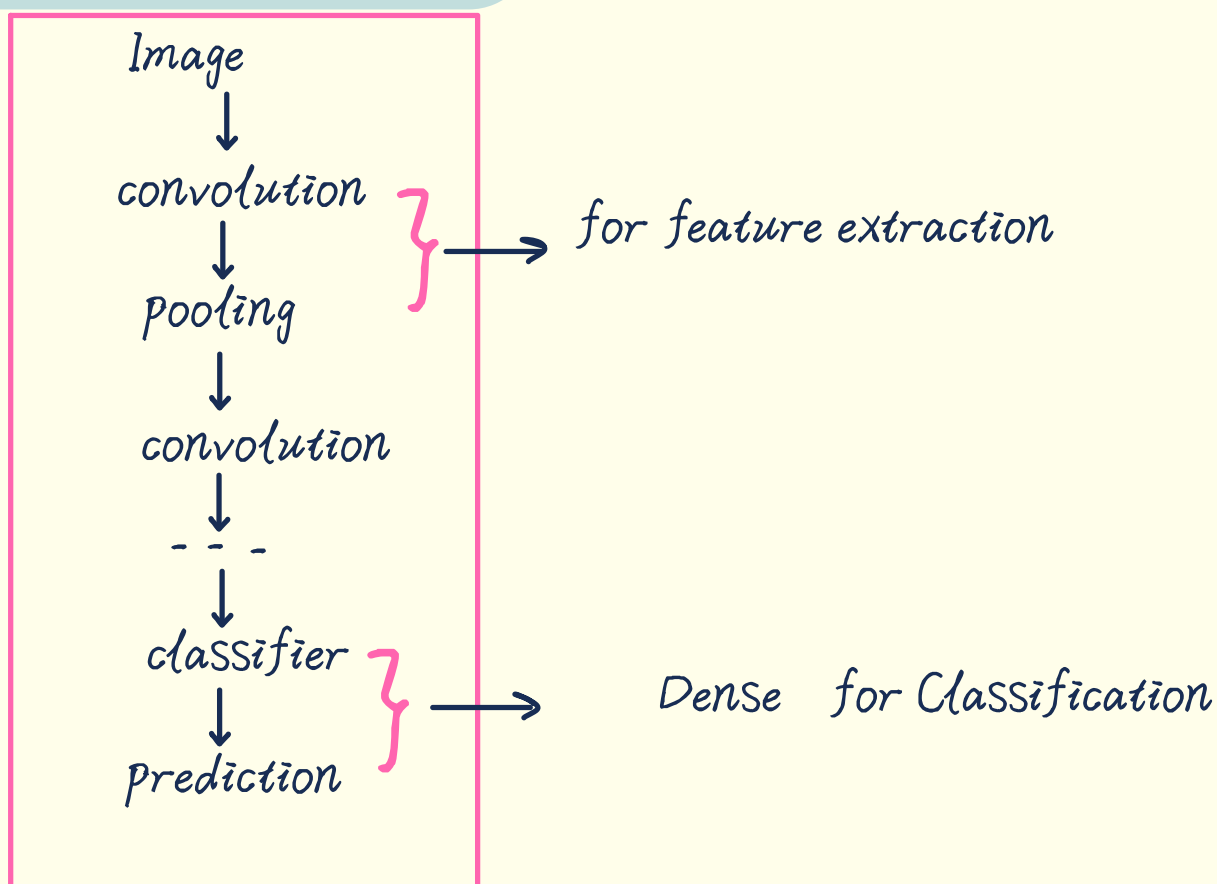
In my notebook I use Maxpooling 2DC)

↳ It takes a small area, most often 2x2 and retains the greatest value

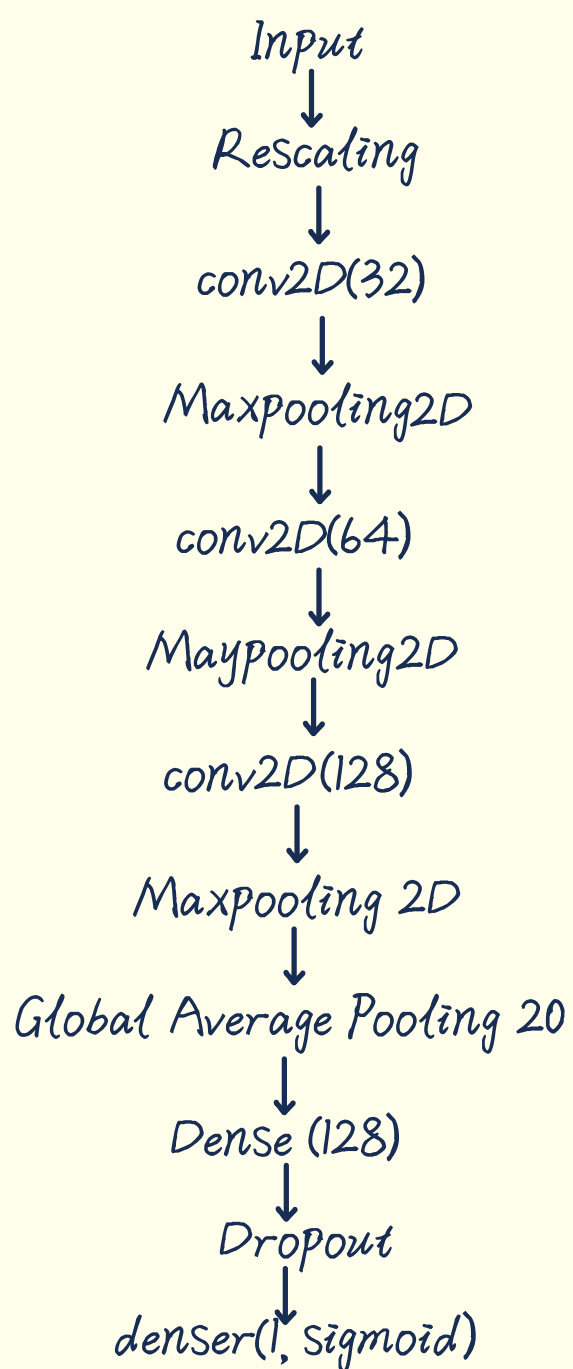


pooling is important because it reduce the amount of data that moves to the next layer thereby reducing calculations and memory , maintaining stronger activation and making the network less sensitive to some small changes in the location of the features

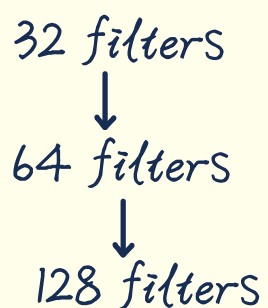
## 2. CNN Architecture



So when I build the model like:-

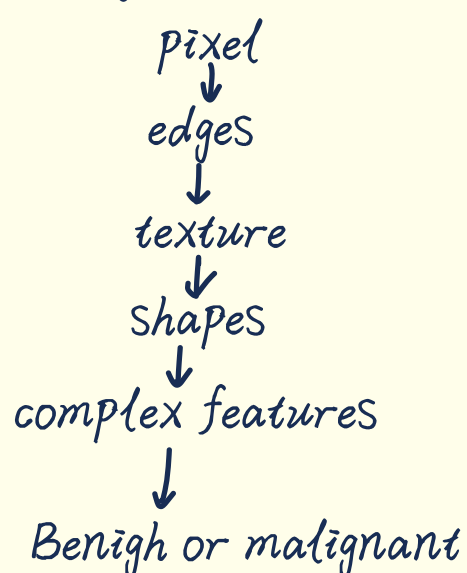


with layers



why the CNN Architecture important?

because the first layer doesn't important  
to learn immediately the meaning of malignant  
Learning is gradual

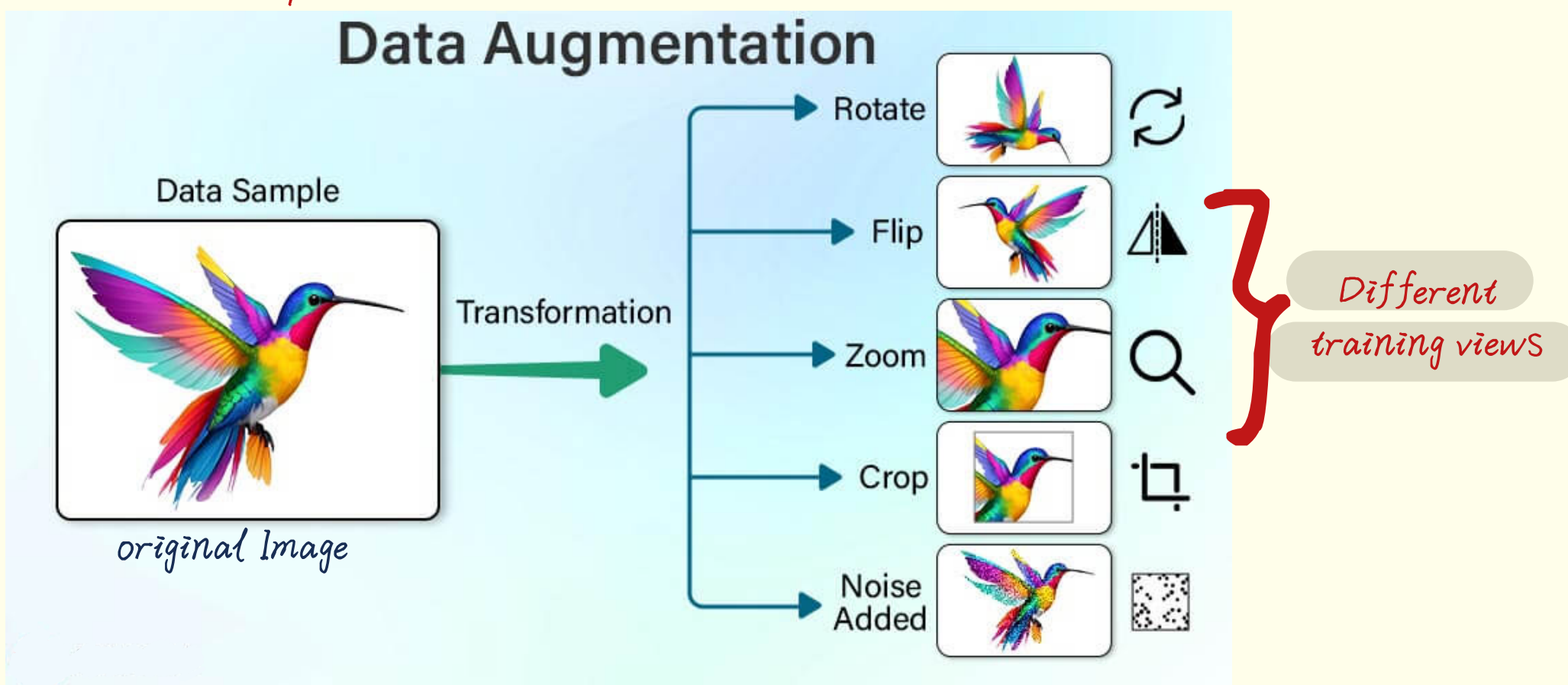


### 3. Data Augmentation

Data Augmentation creates realistic variations of training images without collecting new images



For example



Note :- Augmentation is applied only to training images, not test images

why?

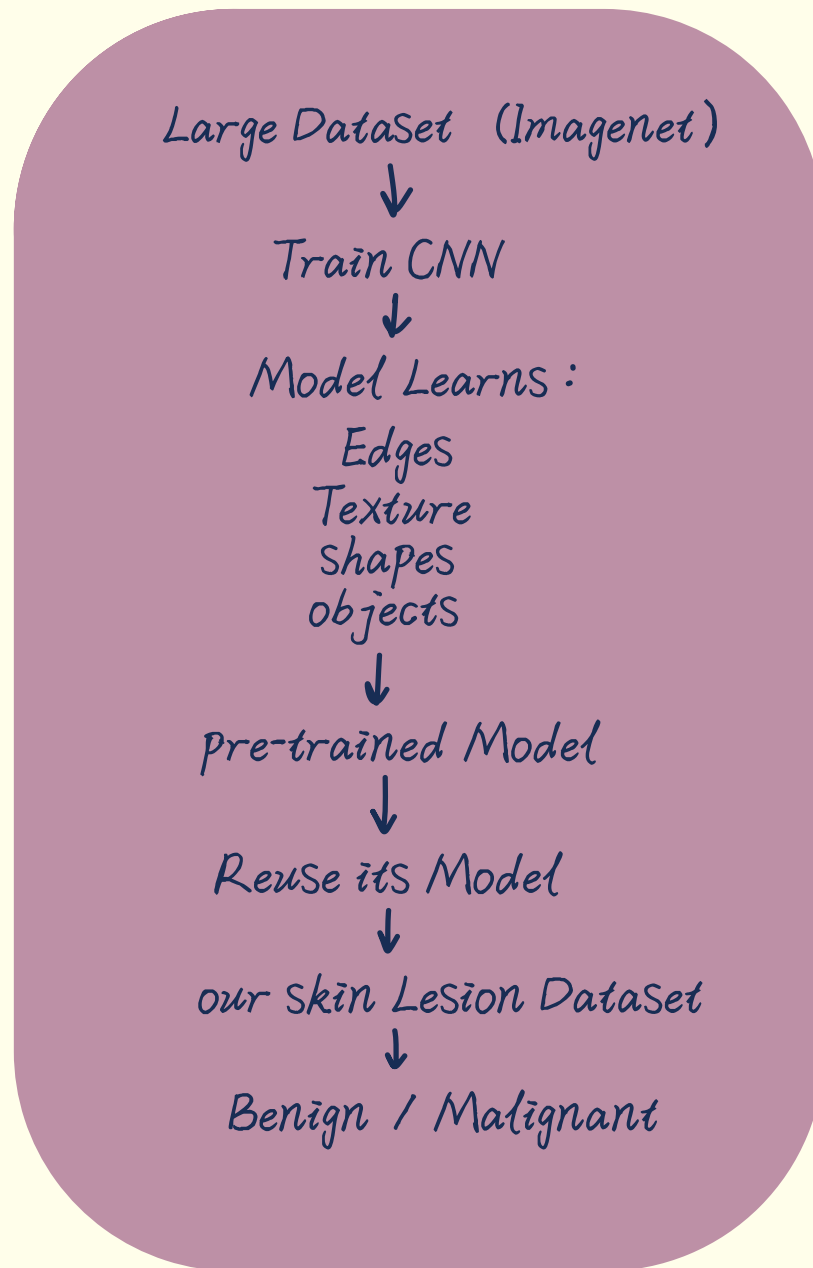


without Augmentation  
- Same image again and again  
↳ Easier to memorize

with Augmentation  
- slightly different images  
↳ Model Learns Stronger patterns

## 4. Transfer Learning

Transfer learning means reusing a model that has already learned useful visual features from a large dataset



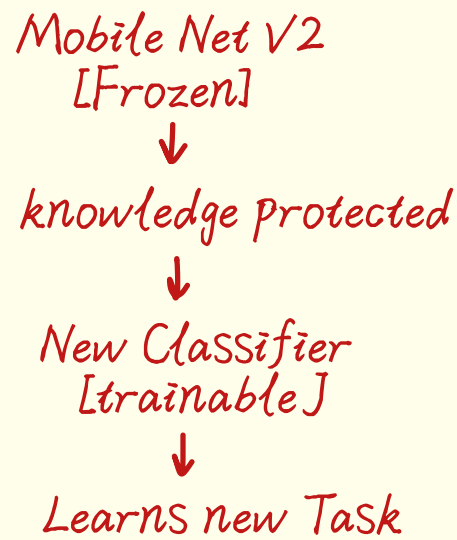
why transfer Learning?

Less Data  
+  
less training Time  
+  
useful pre-trained features

} = Stronge Starting point

## 5. Freezing & Fine Tuning

**Freezing** : means keeping the pre-trained backbone weights unchanged



In my project first freeze Mobile Net and train only the new classifier

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## 6. Fine Tuning

Fine Tuning means unfreezing some upper layers of the pre-trained model and training them carefully on our dataset



**Note** : Do Not fine-tune too early with a large Learning rate because it may destroy useful pre-trained knowledge



# Freezing vs Fine Tuning

|               | Freezing             | Fine-Tuning                          |
|---------------|----------------------|--------------------------------------|
| Backbone      | Fixed                | Some layers train                    |
| Learning Rate | Normal               | Very small                           |
| Goal          | Train new classifier | Adapt learned features               |
| Risk          | Low                  | Can damage weights if too aggressive |